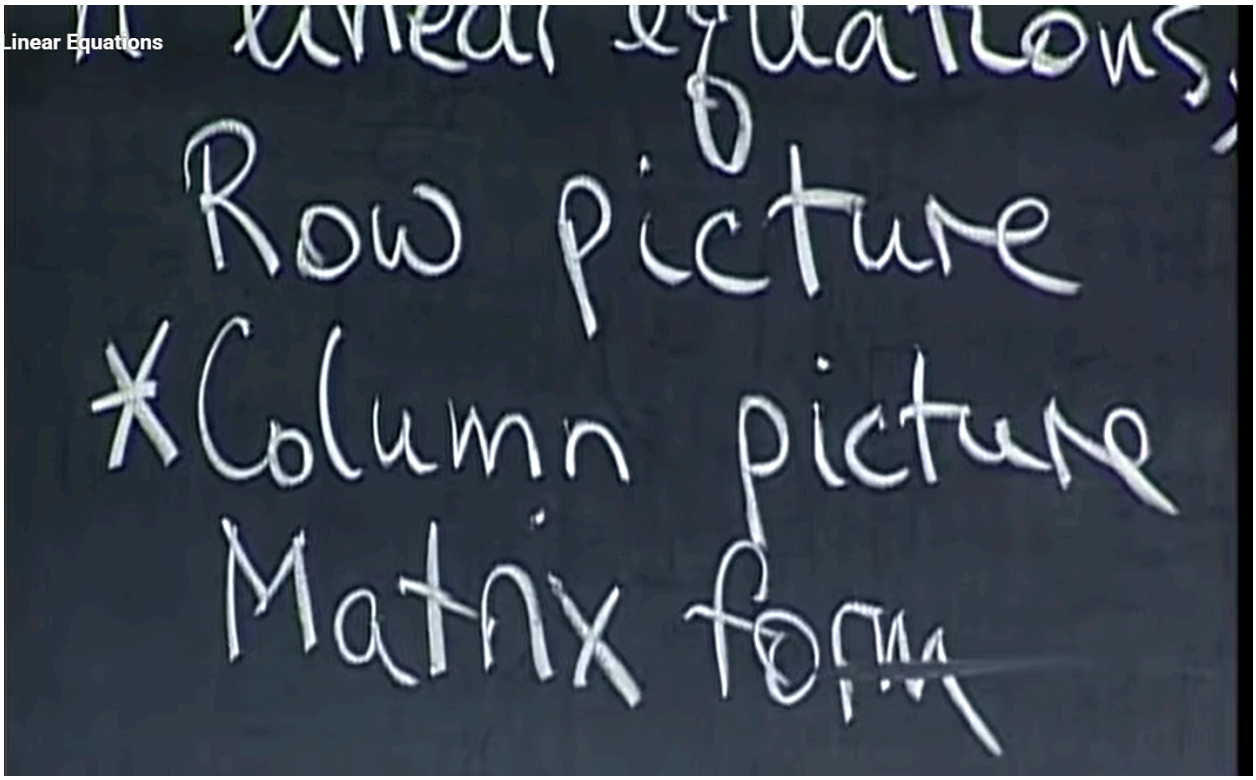
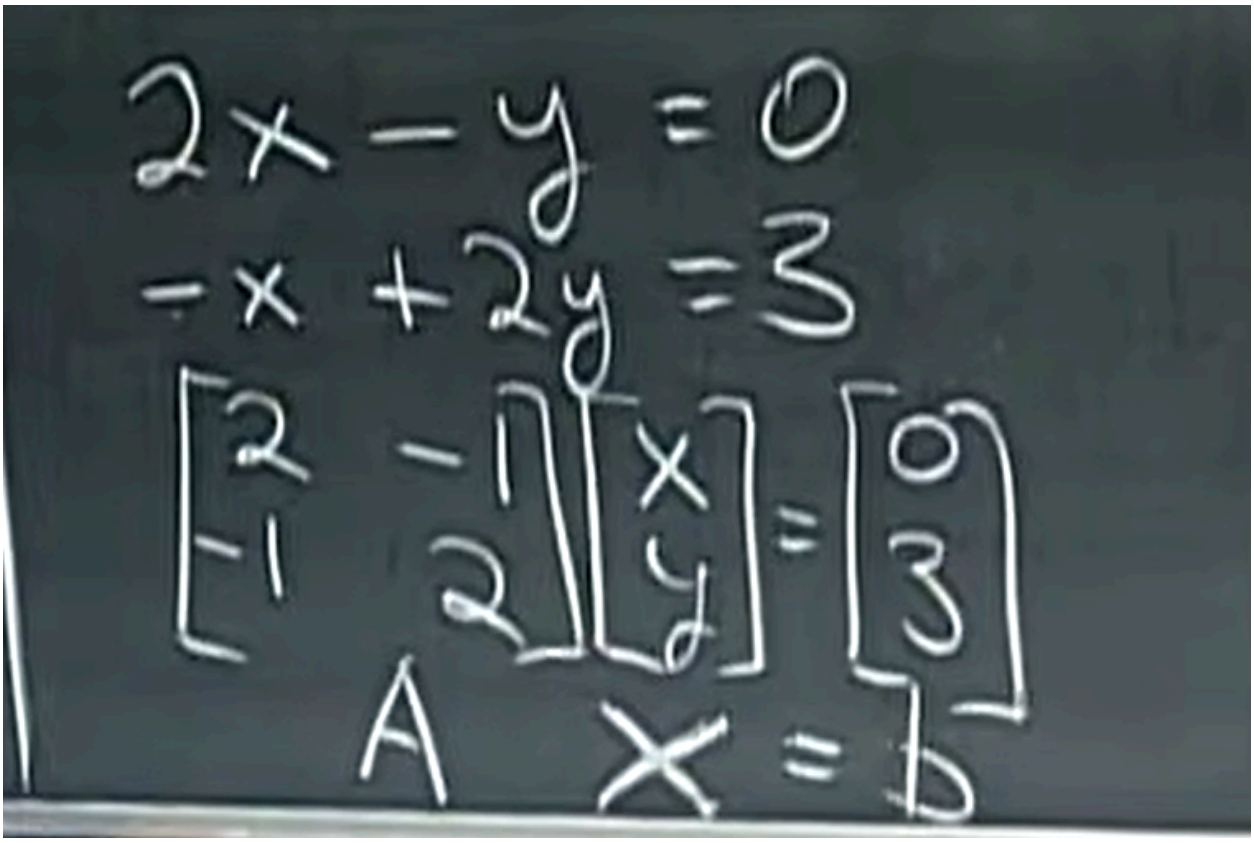
Linear Equations





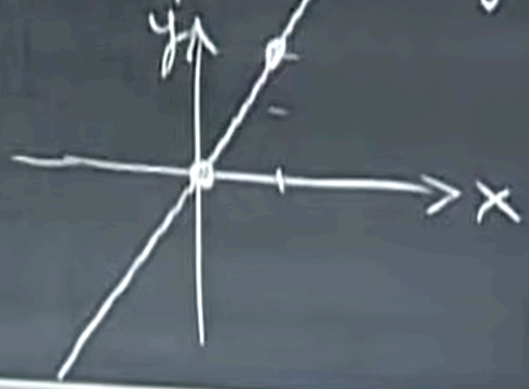
$2x - y = 0$

$-x + 2y = 3$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$A \quad X = b$

Row picture $2x - y = 0$



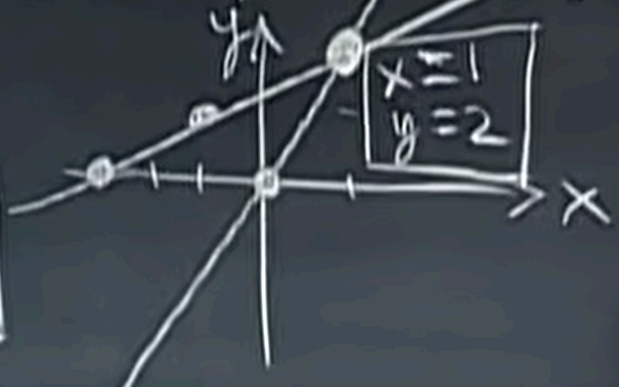
$2x - y = 0$

$-x + 2y = 3$

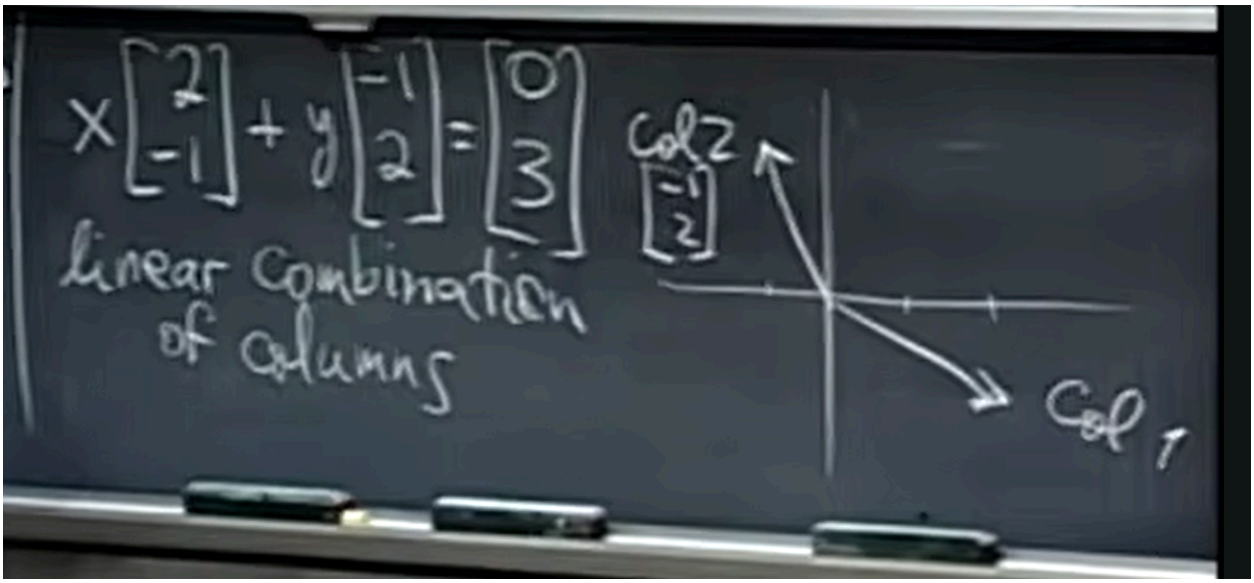
$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

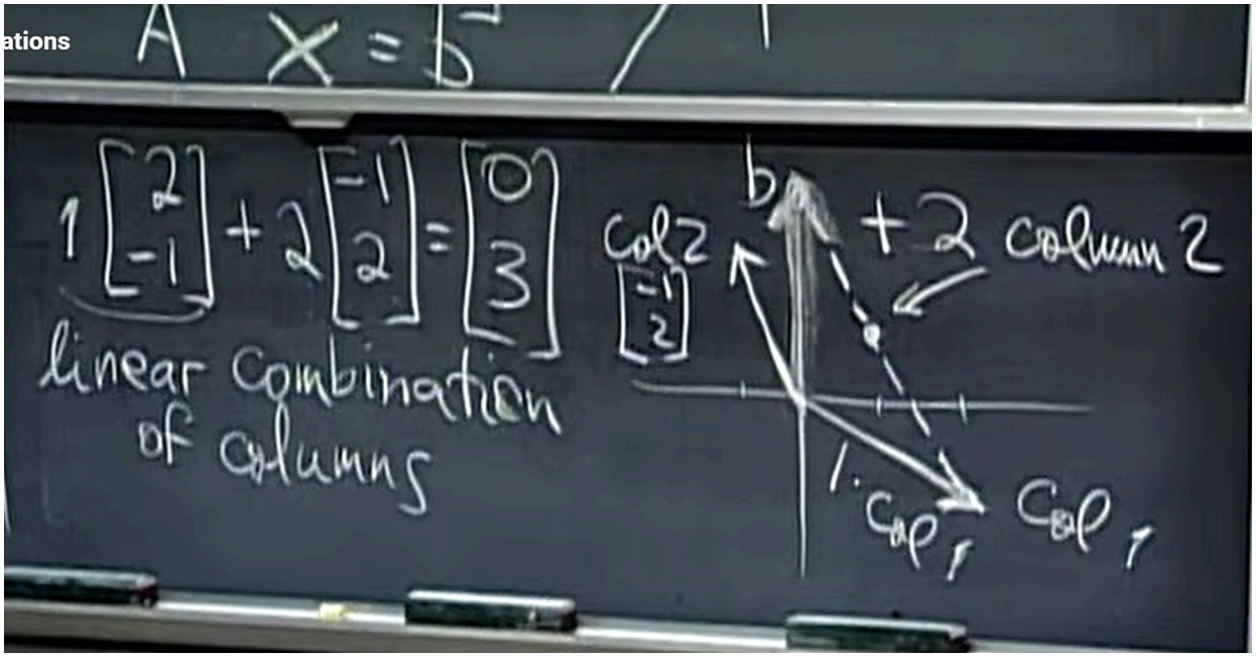
$A \quad X = b$

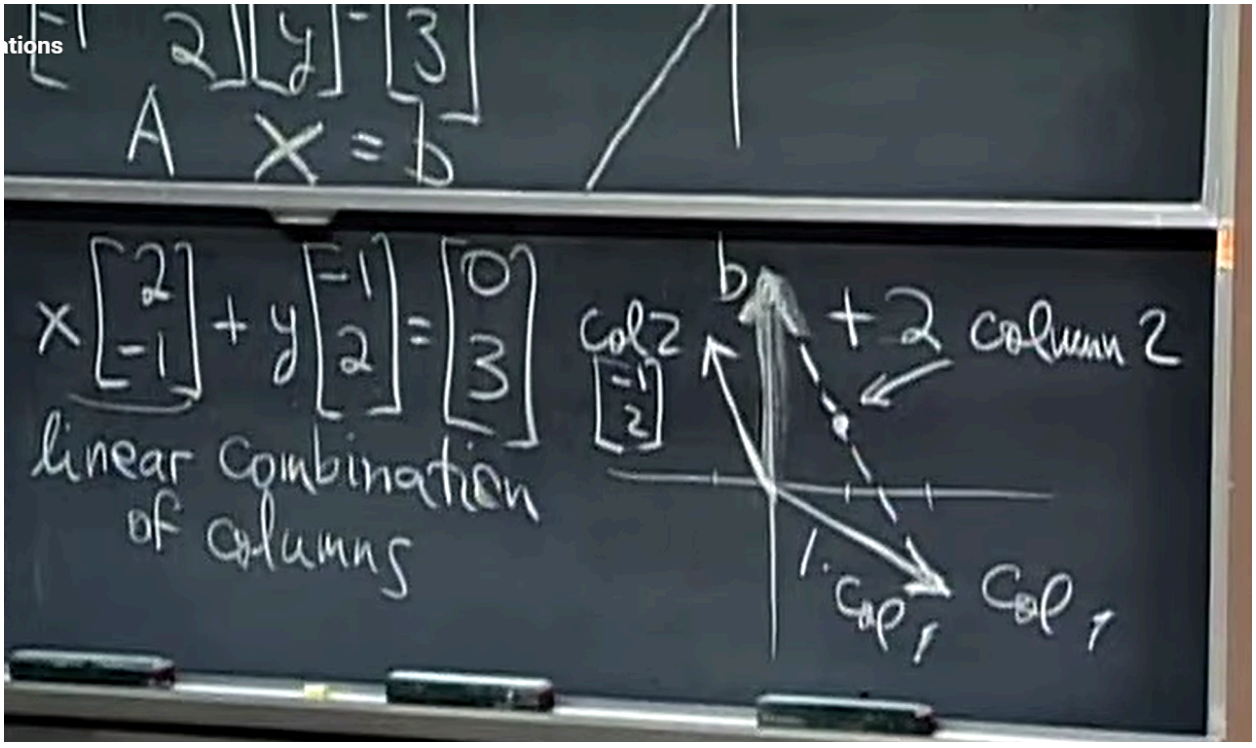
Row picture $2x - y = 0$

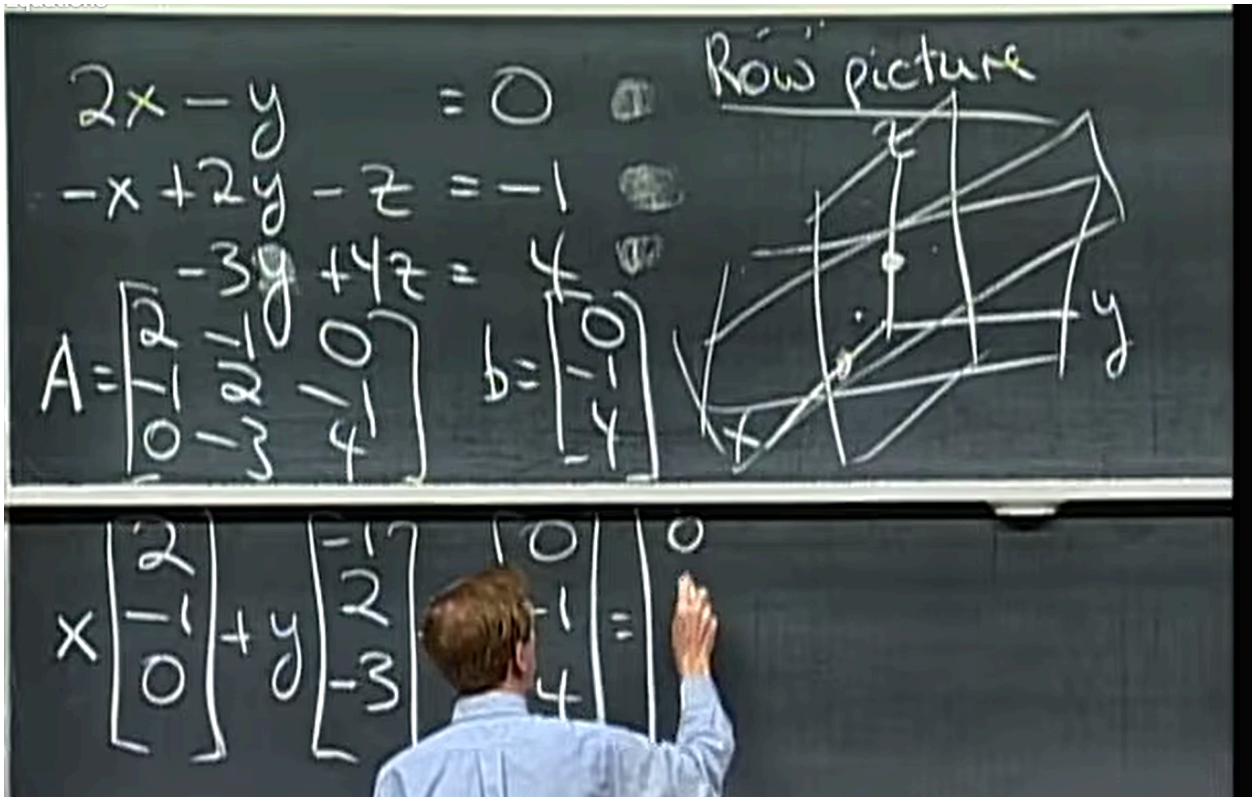


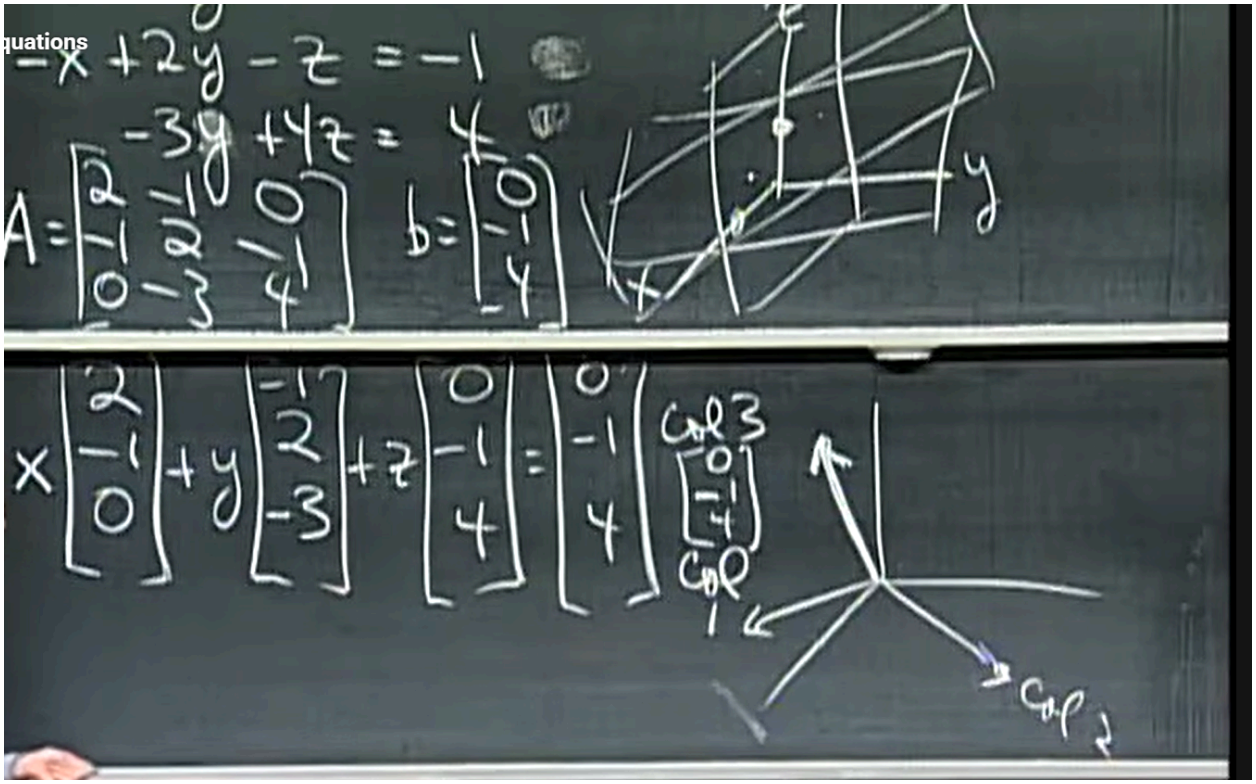
$$x\begin{bmatrix} 2 \\ -1 \end{bmatrix} + y\begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

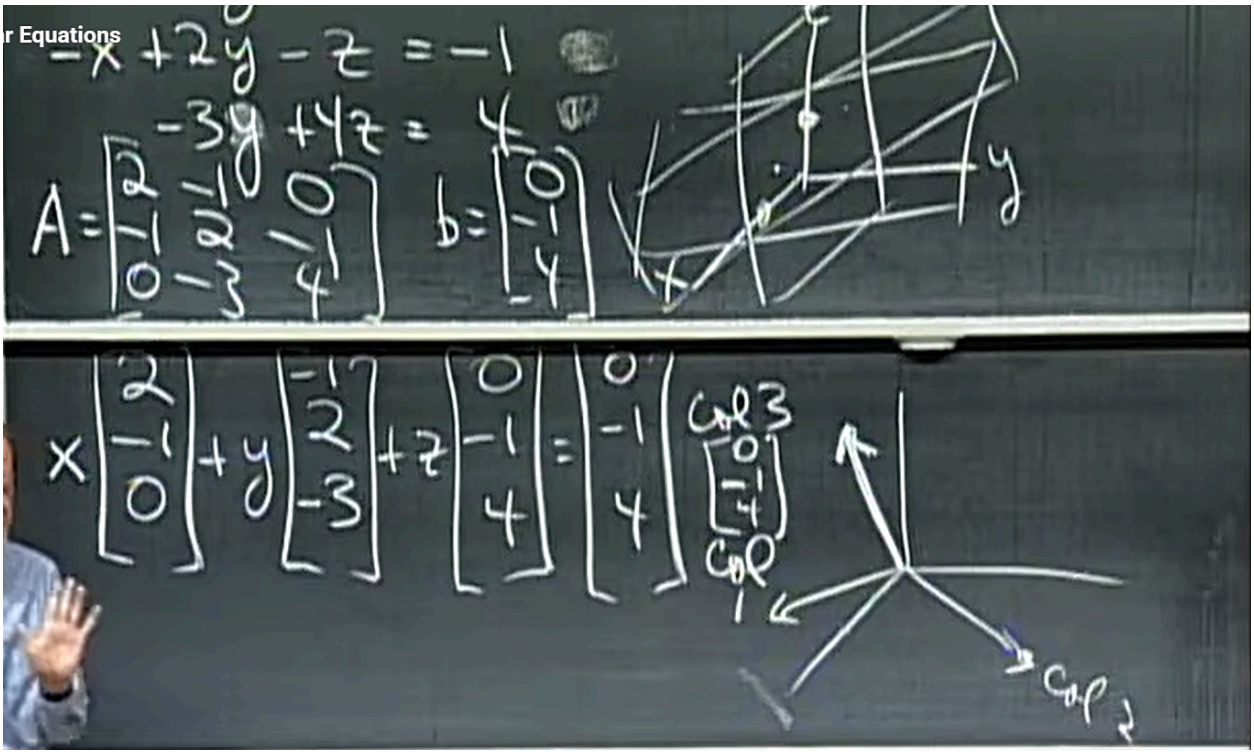

$x\begin{bmatrix}2\\-1\end{bmatrix}+y\begin{bmatrix}-1\\2\end{bmatrix}=\begin{bmatrix}0\\3\end{bmatrix}$
linear combination of columns
col 2
col 1

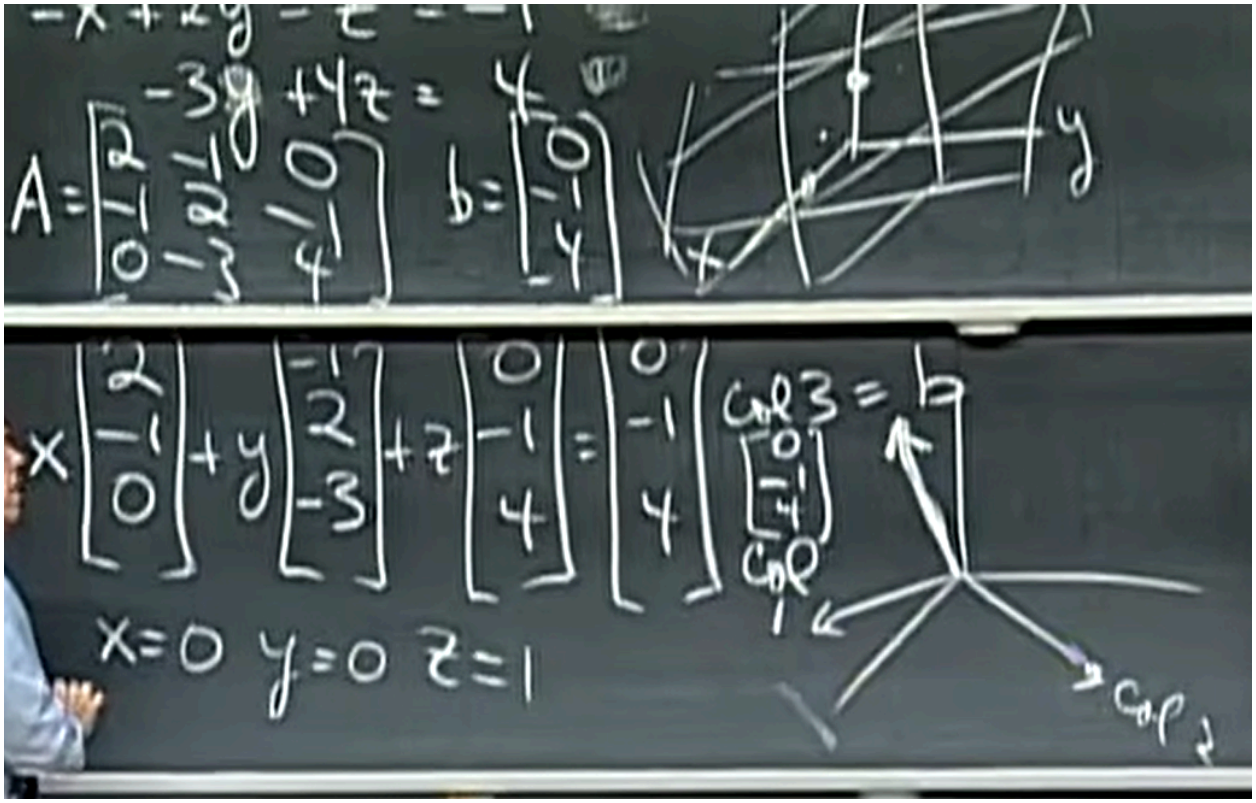

ations
$A\ X = b$
$1\begin{bmatrix}2\\-1\end{bmatrix}+2\begin{bmatrix}-1\\2\end{bmatrix}=\begin{bmatrix}0\\3\end{bmatrix}$
linear combination of columns
b
+2 column 2
col 2
1 · col 1
col 1

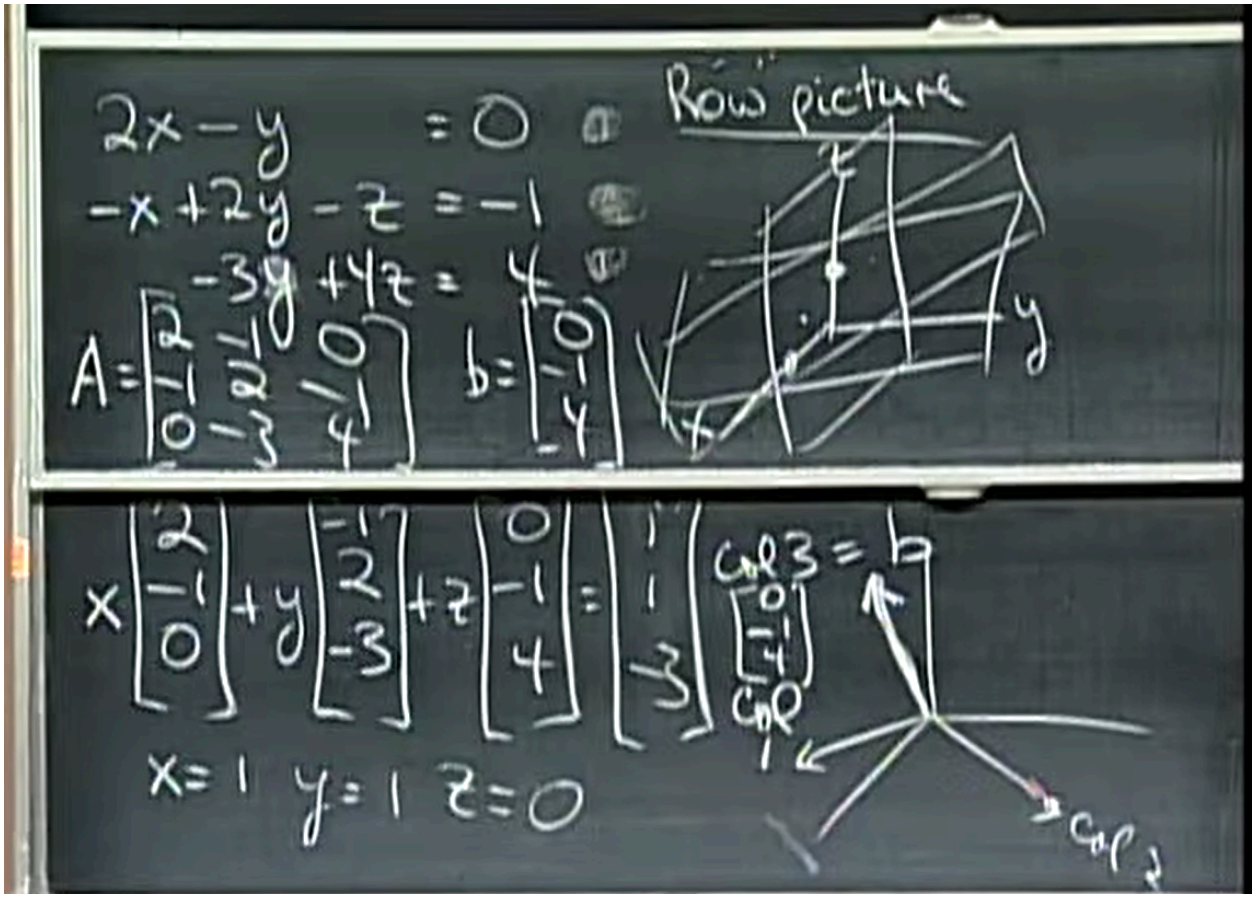

Ax = b
x [2 -1] + y [-1 2] = [0 3]
linear combination of columns
b
+2 column 2
col 2
1 · col 1
col 1


2x - y = 0
-x + 2y - z = -1
-3y + 4z = 4
A = [2 -1 0; -1 2 -1; 0 -3 4]
b = [0 -1 4]
Row picture
x [2 -1 0] + y [-1 2 -3] = [0 -1 4]


quations
x + 2y - z = -1
-3y + 4z = 4
A = [1 2 -1; 1 2 0; 0 -3 4]
b = [0; -1; 4]
x [1; 1; 0] + y [2; 2; -3] + z [-1; 0; 4] = [0; -1; 4]
Col 3
Col 1
Col 2


r Equations
x + 2y - z = -1
-3y + 4z = 4
A = [1 2 -1; 1 2 0; 0 -3 4]
b = [0; -1; 4]
x [1; 1; 0] + y [2; 2; -3] + z [-1; 0; 4] = [0; -1; 4]
Col 3
Col 1
Col 2


A =
b =
y
x
y
z
Col 3 = b
Col 1
Col 2
x=0 y=0 z=1


2x − y = 0
−x + 2y − z = −1
−3y + 4z = 4
A = [2 −1 0; −1 2 −1; 0 −3 4]
b = [0; −1; 4]
Row picture
z
y
x
x [2; −1; 0] + y [−1; 2; −3] + z [0; −1; 4] = [0; −1; 4]
col 3 = b
col 2
col 1
x = 1 y = 1 z = 0

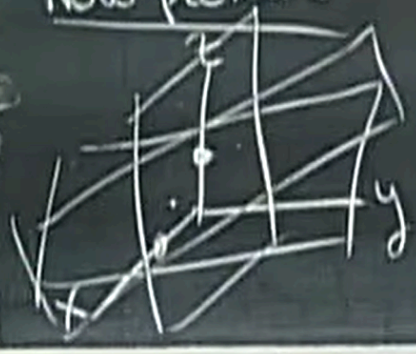
$2x - y = 0$ (1)

$-x + 2y - z = -1$ (2)

$-3y + 4z = 4$ (3)

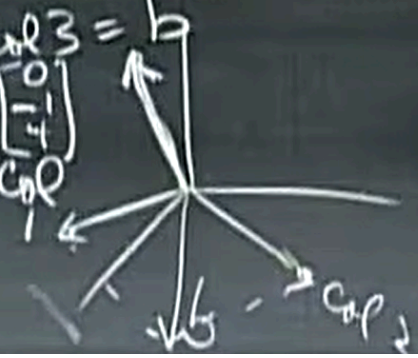
$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix}$ $b = \begin{bmatrix} 0 \\ -1 \\ -4 \end{bmatrix}$

Row picture



$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ -4 \end{bmatrix}$

$x = 1 \quad y = 1 \quad z = 0$

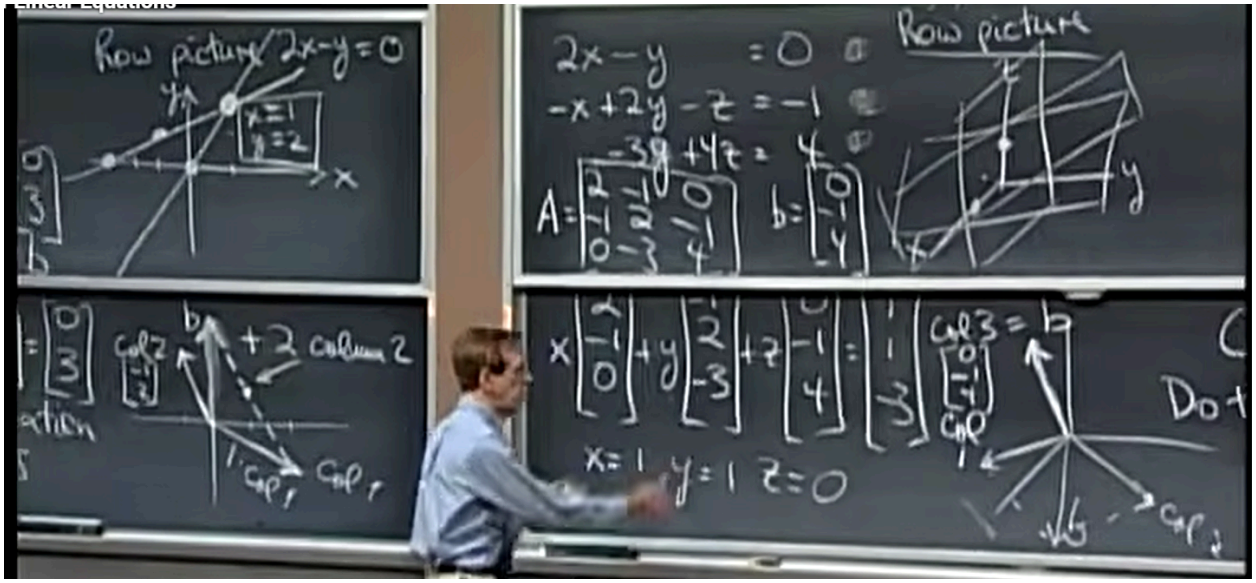


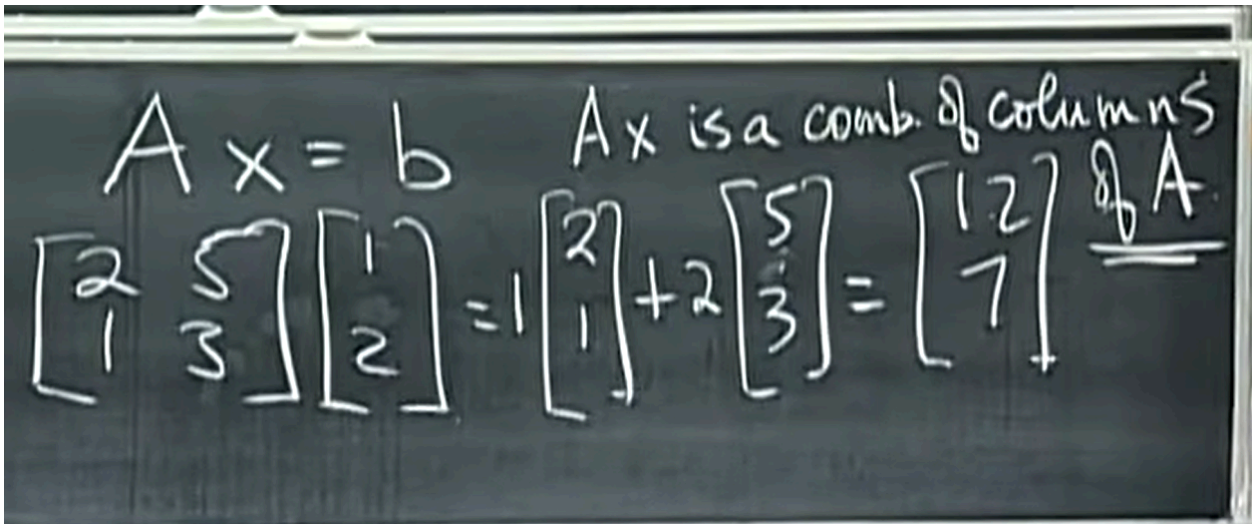
Can I solve $Ax = b$ for every b?

Do the linear combs. of the columns fill 3-D space?

For this A, answer is YES




Row picture
2x − y = 0
−x + 2y − z = −1
−3y + 4z = 4
Row picture
x = 1
y = 1
z = 0
+2 column 2


Ax = b
Ax is a comb. of columns of A
[2 5; 1 3][1; 2] = 1[2; 1] + 2[5; 3] = [12; 7]